Morning Paper Report - Tuesday, July 21, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Geometry of naturalistic object representations in recurrent neural network models of working memory

Xiaoxuan Lei, Takuya Ito, P. Bashivan
Neural Information Processing Systems, 2024

Source: Semantic Scholar; matched terms: working memory, neural network, neural networks, recurrent neural network

Link: https://www.semanticscholar.org/paper/51aea43ac09b9495b4d0c3c128e68fbaf99c5e48

Goal: Working memory is a central cognitive ability crucial for intelligent decision-making

Method: Recent experimental and computational work studying working memory has primarily used categorical (i.e., one-hot) inputs, rather than ecologically relevant, multidimensional naturalistic ones

Results: As a result, an understanding of how naturalistic object information is maintained in working memory in neural networks is still lacking

Conclusion: Our findings indicate that goal-driven RNNs employ chronological memory subspaces to track information over short time spans, enabling testable predictions with neural data.

2. A recurrent neural network model of prefrontal brain activity during a working memory task

Emilia P Piwek, M. Stokes, C. Summerfield
bioRxiv, 2022

Source: Semantic Scholar; matched terms: working memory, neural network, neural networks, recurrent neural network

Link: https://www.semanticscholar.org/paper/709e391bf64dc5c1ce6cdbb3fa48de385534751d

Goal: When multiple items are held in short-term memory, cues that retrospectively prioritise one item over another (retro-cues) can facilitate subsequent recall

Method: We extend these findings by analysing the learning dynamics and connectivity patterns of the RNNs, as well as the behaviour of models trained with probabilistic cues, allowing us to make predictions for future studies

Results: We found that the orthogonal-to-parallel geometry transformation observed in the macaque LPFC emerged naturally in RNNs trained to perform the task

Conclusion: Interestingly, the parallel geometry only developed when the cued information was required to be maintained in short-term memory for several cycles before readout, suggesting that it might confer robustness during maintenance

3. Redundancy Repartitions Capacity and Enhances Recall in Visual Working Memory: The Sample-Size Model of Neural Activation

Authors unavailable
2026

Source: OSF/PsyArXiv; matched terms: visual working memory, working memory, precision, modeling

Link: https://osf.io/preprints/psyarxiv/8x4wg_v2/

Goal: Visual working memory (VWM) has a limited capacity of around four items, yet recall is also improved by structure in the memory array

Method: We reconcile these views by testing how color redundancy – a low-level structural regularity – alters recall precision in a continuous color-reproduction task using a between-subjects (21 participants) and small-N within-subjects design (5 participants)

Results: Visual working memory (VWM) has a limited capacity of around four items, yet recall is also improved by structure in the memory array

Conclusion: These individual differences would be masked by data aggregation Our results motivate the use of individual-level modeling when theorizing about nuanced relationships, such as exists between attention and VWM.

4. Distributed networks for auditory memory differentially contribute to recall precision

Sung-joo Lim, C. Thiel, B. Sehm, L. Deserno, J. Lepsien, J. Obleser
bioRxiv, 2021

Source: Semantic Scholar; matched terms: working memory, fidelity, precision, modelling

Link: https://www.semanticscholar.org/paper/8752cd8018730af25d5593a6d149b0c4715166d1

Goal: Re-directing attention to objects in working memory can enhance their representational fidelity

Method: The present fMRI experiment leverages psychophysical modelling and multivariate auditory- pattern decoding as behavioral and neural proxies of mnemonic fidelity

Results: However, among these regions retaining auditory memory objects, neural fidelity in the left STS and its enhancement through attention-to-memory best predicted individuals' gain in auditory memory recall precision

Conclusion: However, among these regions retaining auditory memory objects, neural fidelity in the left STS and its enhancement through attention-to-memory best predicted individuals' gain in auditory memory recall precision Our results demonstrate how functionally discrete brain regions differentially contribute to the attentional enhancement of memory representations.

5. Reviewer #1 (Public review): Neural dynamics of visual working memory representation during sensory distraction

Authors unavailable
2025

Source: Crossref; matched terms: visual working memory, working memory representation, working memory

Link: https://doi.org/10.7554/elife.99290.3.sa1

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.